

Muhammad Usman

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EDUCATION

National University of Computing And Emerging Sciences (FAST)

Bachelor of Science in Software Engineering

Islamabad, Pakistan

Aug. 2022 – May 2026

EXPERIENCE

Lab Demonstrator / Teaching Assistant

National University of Computing And Emerging Sciences (FAST)

Islamabad, Pakistan

Aug. 2023 – Present

- Mentored 40+ undergraduate students in fundamental Computer Science courses (Data Structures & OOP), reinforcing core programming concepts.
- Conducted code reviews and debugging sessions to assist students in troubleshooting complex C++ and Java assignments.

ACADEMIC & TECHNICAL PROJECTS

Libaayah E-Commerce Store | [Live Site](#) | [GitHub Repository](#)

React, Clerk Auth, Firebase/Firestore, Node.js, Express, Shopify Storefront API, GraphQL |

- Developed a production-ready full-stack e-commerce platform integrating Shopify Storefront GraphQL API with a custom authentication and data-persistence layer.
- Built a responsive React SPA with real-time cart sync, wishlist, order history, reusable component architecture, and global state via custom Context providers.
- Implemented secure user authentication using Clerk (JWT session verification) with synchronized cart + profile data stored in Firebase/Firestore.
- Developed Node.js/Express backend for cart persistence, order retrieval, webhook processing, and Shopify checkout workflows with HMAC-verified webhooks.

Secure End-to-End Encrypted Messaging System | [GitHub Repository](#)

React, Web Crypto API, IndexedDB, Node.js, Express, MongoDB, Socket.io

- Built a secure real-time messaging platform with true end-to-end encryption where the server cannot decrypt user messages or files (zero-knowledge architecture).
- Implemented custom ECDH key exchange with ECDSA-signed messages, HKDF-SHA256 key derivation, and replay-attack protection using nonces, timestamps, and sequence counters.
- Developed client-side crypto layer using Web Crypto API (AES-256-GCM for messages/files) with secure key storage in IndexedDB and password-based encryption.
- Built Node.js/Express backend for encrypted message transport, JWT authentication, metadata storage in MongoDB, and real-time communication via Socket.io.

VoxaAI — AI-Powered Audio Enhancement Platform [GitHub Repository](#)

React Native, Expo, Node.js, FastAPI, Python, MongoDB, Docker

- Developed a mobile-first AI audio enhancement platform with microservices architecture (Gateway, User Service, AI Service) and real-time processing progress tracking.
- Implemented AI pipeline: Faster-Whisper speech transcription, Gemini 2.0 grammar enhancement, and zero-shot voice cloning using IndexTTS2 on GPU infrastructure.
- Built secure backend services with JWT authentication, RESTful APIs, file processing workflows, and metadata storage using MongoDB Atlas.
- Developed cross-platform React Native app (Expo) with recording, waveform playback, authenticated user profiles, and cloud-synced processing history.

TECHNICAL SKILLS

Languages: JavaScript (ES6+), TypeScript, Python, C/C++

Frontend: React 19, React Native (Expo), Vite, React Router DOM, Tailwind CSS, Context API, LocalStorage Sync

Backend & Integrations: Node.js, Express.js, FastAPI, Firebase Admin SDK, Firestore, Clerk Auth, Shopify Storefront GraphQL API, REST APIs, Webhooks, Git, GitHub